

Qatar-1b

R-2737

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_140829 · created 2026-09-02T01:07:00+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456898.67519 +/- 0.00036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.117 +/- 0.013
Transit depth (Rp/Rs)^2	1.38 +/- 0.31 [%]
Orbital Inclination (inc)	85.04 +/- 1.4
Airmass coefficient 1 (a1)	0.983 +/- 0.01
Airmass coefficient 2 (a2)	0.013 +/- 0.053
Scatter in the residuals of the lightcurve fit is	1.03 %
Best Comparison Star	None
Optimal Aperture	3.68
Optimal Annulus	11.08
Transit Duration (day)	0.0705 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.117 ± 0.013 vs 0.1463 (deviation 2.23σ)
Duration fit vs published	0.0705 d vs 0.0692 d (deviation 0.2σ)
Mid-time O–C	0.9 min
Depth SNR	8.9
Residual scatter	1.03 %

Light curve

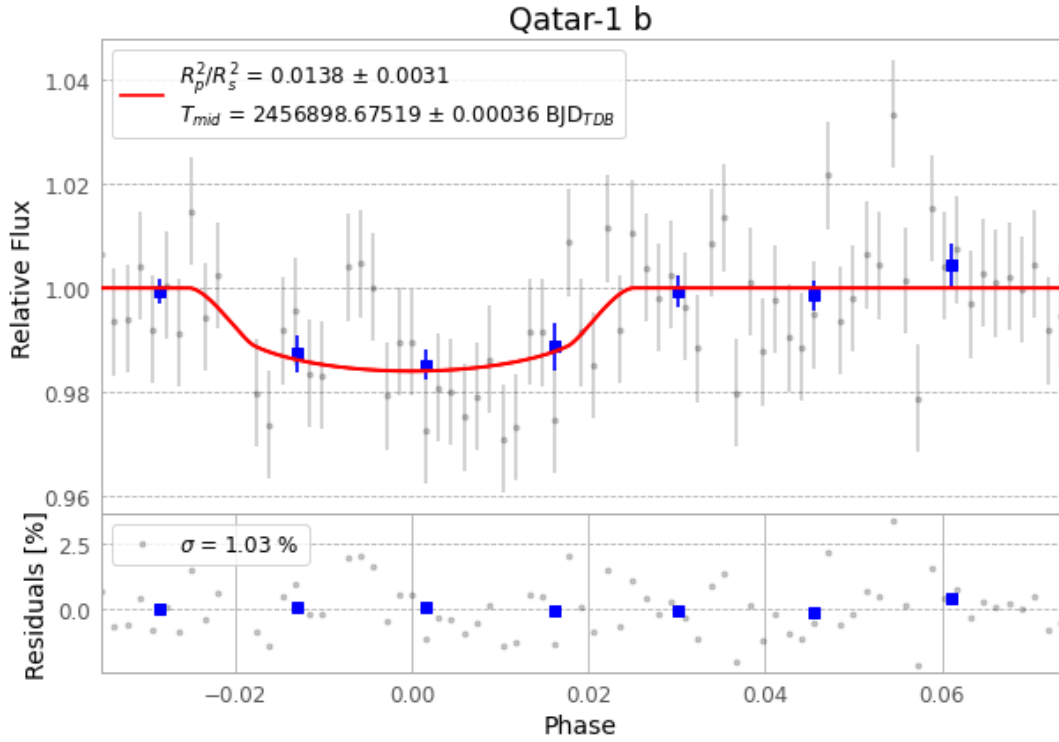


Figure 1 — FinalLightCurve_Qatar-1 b_2014-08-28.png (SHA-256 78cd2d767ff212e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [302, 253], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ec9cc02d937058d1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

75 FITS frames (49 MB), Qatar-1140829025712.FITS → Qatar-1140829063912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-140829.log (efb21e76064f...), FinalLightCurve_Qatar-1 b_2014-08-28.png (78cd2d767ff...), FinalLightCurve_Qatar-1 b_2014-08-28.csv (2e8e348cd1e5...)

Compute resources

Wall time 438.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 01:07Z.